

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			A: Mitochondrion/mitochondria B: Golgi {body/complex/apparatus} C: Ribosome(s) D: Nuclear pore 2 or 3 correct for 1 mark All correct for 2 marks	2			2		
	(b)	(i)		Nucleus/ E: (contains DNA which) codes for the production of {proteins/polypeptides/sequence of amino acids in a polypeptide}/ transcription/ {pre-processing/ production/ synthesis} of mRNA (1) Nuclear pores/ D: allow {mRNA/rRNA} to leave the nucleus (1) Ribosome(s)/C: carry out translation/ protein synthesis/or description of (1)	3			3		

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1	(b)	(ii)		Any 3 × (1) from: Rough Endoplasmic Reticulum/ RER/ F: transports proteins {through the cell/ through the cytoplasm}/ transports proteins to golgi body/ package proteins into vesicles (1) Golgi body/ B: {Packaging/Modification} of protein/description of/ activation of enzyme (1) (transport) vesicle/ G: transports {proteins/enzymes} to the {cell membrane/ plasma membrane} (1) Exocytosis (of enzymes from the cell) (1)	3			3		
		(iii)		<u>Provide ATP :</u> {for transcription / translation / protein synthesis} (1) <u>exocytosis</u> (1)		2		2		
				Question 1 total	8	2	0	10	0	0